

PRASHANT BANJARE

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Education

National Institute of Technology, Rourkela

November 2022 – Present

Bachelor of Technology in Metallurgical and Materials Engineering (CGPA - 7.47)

Rourkela, Odisha

Relevant Coursework

- Data Structures and Algorithms
- Machine Learning
- Natural Language Processing
- Deep Learning
- Statistics
- Object-Oriented Programming

Work Experience

HealthyBites Express

May 2025 – July 2025

AI/ML Intern

Certificate

- Collaborated with **engineering, medical and product teams** to design a **meal recommendation system**.
- Executed **A/B testing** on **500 users** per model and analyzed user feedback data to enhance model performance.
- Achieved **89%** conversion rate using XGBoost with personalized dietary recommendations (**F1-score: 0.87**).

Projects

Voice Research Paper Explainer using Generative AI

April 2026

FastAPI, React.js, LangChain, FAISS, Groq API, NLP, Tailwind CSS

GitHub

- Extracted text from **100** page PDFs and generated **500**-character chunks with **100** overlap for retrieval.
- Converted research content into **embeddings** and retrieved **top 3** relevant chunks using **FAISS similarity search**.
- Built **2-mode** interaction system with **voice input, text-to-speech output**, and responsive AI chat interface.

Real vs Fake Image Classification Model using Deep Learning

December 2025

Convolutional Neural Network, TensorFlow, FastAPI, REST API

GitHub

- Processed over **5,400** images, resized to **256×256**, rescaled and applied **augmentation** to improve model robustness.
- Implemented **L2-regularized CNN** with **4** dense layers, enhancing generalization for binary image classification.
- Achieved **94.57% accuracy** with **F1-scores of 0.94** for both classes; showcasing excellent image detection.
- Deployed the trained CNN model using FastAPI, enabling real-time image classification through a REST API.

IPL Match Winner Prediction Model using Machine Learning

June 2025

Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, XGBoost, SQL, Streamlit

GitHub

- Processed **756** IPL matches and **179,078** deliveries; performed EDA with **10** visualizations to uncover insights.
- Engineered **10** key features; built XGBoost pipeline with One-Hot Encoding, achieving **F1-score of 0.99**.
- Wrote **10** SQL queries; created **Streamlit UI** and **deployed application** on Render using GitHub for version control.

Certification

Machine Learning from Coursera – DeepLearning.AI & Stanford University

November 2025

Supervised and Unsupervised Machine Learning

Certificate

- Developed expertise by solving regression, classification, clustering, and 3 recommendation system problems.

EDA from IBM

June 2025

Exploratory Data Analysis for Machine Learning

Certificate

- Completed **IBM's EDA** course, acquiring hands-on experience through **10+** hours of content and practical labs.

Technical Skills

- **Languages:** Python, C++, SQL, JavaScript, HTML, CSS
- **ML / DL Libraries:** Scikit-learn, TensorFlow, PyTorch, OpenCV, NumPy, Pandas, Seaborn, Matplotlib
- **Generative AI & NLP:** LangChain, Hugging Face Transformers, RAG, Prompt Engineering
- **Web & API:** FastAPI, React.js, Express.js, Tailwind CSS, REST API, Streamlit
- **MLOps & Deployment:** Docker, Git, GitHub, Render, FastAPI serving, Jupyter Notebook
- **Data & BI Tools:** Power BI, Tableau, Microsoft Excel
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Developer Tools:** VS Code, Bash, Jupyter Notebook

Extracurricular Activity

AASRA

April 2023 – Present

Member

NIT Rourkela

- Mentored **20+** students and coordinated Foundation Day and Little Science Yard events, ensuring smooth execution.